

Practical Assessment 1 – CDS 3643 (202520)

25% Weight Week 7

Duration: 60 Minutes

Mode: Open Book

Supporting Material: Syntax Sheet

Exam Room: Computer Lab with FWA Practical exam setting

Printed Material: Not Allowed

Calculator: Not Allowed

Practical Assessment 1 Topics:

- CLO1: Demonstrate an understanding of the concepts of time series data and how languages such as R or Python can be used to perform time series analysis
- CLO2: Apply descriptive analysis of time series data to provide insights for time-series analysis

Practical Assessment 1 Topic Weights and Number of Questions:

CLO	Questions	Marks
CLO1	Q1, Q2	20 (10 + 10)
CLO2	Q3, Q4	20 (10 + 10)
Total		40

Practical Exam Dataset: sales.csv

The dataset contains the hourly sales on e-commerce platform.

CLO1 Q1 (10 Marks)

You are provided with the sales dataset, which contains hourly sales information.

Tasks:

1. Import the required Python libraries for time series analysis.
2. Load the dataset into a Pandas DataFrame.
3. Display the first few rows and list the column names.
4. Convert the Date column into a proper datetime format.
5. Set the Date column as the time index of the DataFrame.
6. Check for invalid or missing values (in both columns date and sales) and handle them appropriately.
7. Select sales as the target variable and create a daily or monthly aggregated time series.

Rubric:-

Task	Marks
Library import & data loading	2
Inspection & datetime conversion	3
Time index setting & cleaning	3
Aggregation to time series	2
Total	10

CLO1 Q2 (10 Marks)

Using the aggregated daily time series created in Q1, perform the following analysis:

Tasks:

1. Plot the time series using a line chart with proper title and axis labels.
2. Apply additive decomposition to the time series and display its components.
3. Apply multiplicative decomposition to the same time series and display its components.
4. Compare the additive and multiplicative decomposition results.
5. Briefly state which decomposition model is more suitable for this dataset and justify your answer in 1–2 lines.

Rubrics: -

Task	Marks
Time series visualization	2
Additive decomposition	3
Multiplicative decomposition	3
Comparison & justification	2
Total	10

CLO2 Q3 (10 Marks)

Using the prepared daily time series from the dataset:

Tasks:

1. Convert the aggregated data into a time-series DataFrame with a proper date/time index.
2. Apply weekly Simple Moving Average (SMA) using the appropriate window size.
3. Plot both the original and SMA-smoothed time series on a single graph, including:
 - a. Title
 - b. Axis labels
 - c. Legend
4. Briefly state (1–2 lines) the purpose of applying SMA in time-series analysis.

Rubrics: -

Task	Marks
Time-series DataFrame creation	2
SMA application (single window)	3
Multiple series plotting	3
Purpose explanation	2
Total	10

CLO2 Q4 (10 Marks)

Using the daily same time series:

Tasks:

1. Apply Holt's method to capture the level and trend of the time series.
2. Apply the Holt-Winters method to capture level, trend, and seasonality.
3. Plot the following on a single graph:
 - a. Original time series
 - b. Holt's smoothed series
 - c. Holt-Winters smoothed series
4. Briefly compare the smoothing results and state which method better represents the data, with a short justification (2–3 lines).

Rubric:

Task	Marks
Holt's method implementation	3
Holt-Winters implementation	3
Combined visualization	2
Interpretation & justification	2
Total	10